

Chapter 10

Linear PDE with constant coefficient homogeneous

(corresponds to chapter 12 of the Chinese reference book)

线性齐次常系数偏微分方程

Example $u_{xx} - 2u_{xy} - 3u_{yy} = 0$

$u_1(x, y)$ and $u_2(x, y)$ are solutions

$au_1 + bu_2$ is also the solution.

homogeneous : the order of derivative is fixed.

Example $u_{xxx} + 5u_{xyy} + 2u_{xxy} - u_{yyy} = 0$

$$D_x u = u_x$$

$$D_x^2 u = u_{xx}$$

$$D_x^2 u = u_{xx}$$

$$D_x^3 u = u_{xxx}$$

$$D_y u = u_y$$

$$D_x D_y u = u_{xy}$$

$$u_{xx} - 2u_{xy} - 3u_{yy} = 0$$

$$D_x^2 u - 2D_x D_y u - 3D_y^2 u = 0$$

$$(D_x^2 - 2D_x D_y - 3D_y^2)u = 0$$

$$(D_x - 3D_y)(D_x + D_y)u = 0$$

$$(D_x + D_y)u = u_x + u_y$$

$$\begin{aligned} [D_x(D_x + D_y) - 3D_y(D_x + D_y)]u &= u_{xx} + u_{xy} + D_y]u = u_{xx} + u_{yy} - 3(u_{xy} + u_{yy}) \\ &= u_{xx} - 2u_{xy} - 3u_{yy} = 0 \end{aligned}$$

Now this can be solved.

$$\begin{aligned}
 (D_x - 3D_y)(D_x + D_y)u &= 0 \\
 (D_x - 3D_y)u &= 0 \quad \text{or} \quad (D_x + D_y)u = 0 \\
 \begin{cases} u_x + u_y = 0 \\ u_x - 3u_y = 0 \end{cases}
 \end{aligned}$$

$u_x + u_y = 0$ we can get $\phi(y-x)\phi(x,y)(-1) + \phi'(y-x) \cdot 1 = 0$

$u_x - 3u_y = 0$ we can get $\psi(3x+y)\psi'(3x+y) \cdot 3 - 3\phi'(3x+y) = 0$

So,

$$u_x - \alpha_1 u_y = 0 \rightarrow \phi(\alpha_1 x + y) = 0$$

Example $u_{xx} - 5u_{xy} - 6u_{yy} = 0$

$$\begin{aligned}
 (D_x - bD_y)(D_x + D_y)u &= 0 \\
 \therefore u_x - f(u_y) &= 0 \quad \text{or} \quad u_x + u_y = 0 \\
 \phi(x+y) \quad \text{or} \quad \psi(y-x) &= 0 \Rightarrow u(x,y) = \phi(x+y) + \psi(y-x)
 \end{aligned}$$

Example $(Dx^2 - Dx Dy)u = 0$

$$\begin{aligned}
 Dx(Dx - Dy)u &= 0 \\
 \Rightarrow u_x &= 0 \quad \text{or} \quad (Dx - Dy)u = 0 \\
 \boxed{(A)_x - BD_y)x = 0} & \quad \phi(Bx + Ay) \\
 Dx u = 0, \text{ we } k_0 w A = 1 B = 0 &\Rightarrow \phi(y) \\
 u_{xx} - 3u_x + 2u &= 0 \\
 \Rightarrow e^x, e^{2x} \\
 u_{xx} - 3u_x + 2u &= f(x)
 \end{aligned}$$

If $f(x)$ is special, such as $e^{cx}, \sin ax, p(x)$, we can guess the particular solution.

Example $u_{xx} - 4u_{xy} + 4u_{yy} = 0$

$$\begin{aligned}
 (ux - 2uy) &= 0 \Rightarrow \phi(2x+y) \quad \psi(2x+y) \\
 g(2x+y) &= \phi(2x+y) + \psi(2x+y)
 \end{aligned}$$

Non-zero input

$$\begin{aligned}
 f(x) &= e^{ax+by} \\
 u_{xx} - 3u_{xy} + 2u_{yy} &= e^{ax+by}
 \end{aligned}$$

Let $u = u = Ae^{ax+by}$

$$\begin{aligned}
u_{xx} &= Ae^{ax+by} & u_{xy} &= Aabe^{ax+by} & u_{yy} &= Ab^2e^{ax+by} \\
(Aa^2 - 3Aab + 2Ab^2)e^{ax+by} &= e^{ax+by} \\
A(a^2 - 3ab + 2b^2) &= 1 \\
L(D_x, D_y) &= D_x^2 - 3D_xD_y + 2D_y^2 \\
\therefore A &= \frac{1}{L(a, b)} \\
u_p(x, y) &= \frac{1}{L(a, b)} \cdot e^{ax+by} \quad (L(a, b) \neq 0)
\end{aligned}$$

Example $e^{ax+b} = f(x, y)$

$$\begin{aligned}
Au_x - Bu_y &= 0 & L(a, b) &= 0 & Aa - Bb &= 0 \\
(bD_x - aD_y)u &= e^{ax+by} \\
L(a, b) &= 0 \\
\frac{x}{b}e^{ax+by} &\text{ or } -\frac{y}{a}e^{ax+by}
\end{aligned}$$

Example $(2D_x - 3D_y)(D_x + D_y)u = 5e^{x+y}$

let $v = (2D_x - 3D_y)u$

$$\begin{aligned}
&\begin{cases} (D_x + D_y)v = Se^{x+y} \\ (2D_x - 3D_y)u = v \end{cases} \\
(D_x + D_y)v &= 5e^{x+y} \Rightarrow a = 1, b = 1 \\
L(a, b) &= a + b = 2 \\
v_p(x, y) &= \frac{5}{2}e^{x+y} \\
(2D_x - 3D_y)u &= \frac{5}{2}e^{x+y} \\
L(a, b) &= 2a - 3b = -1 \\
u_p &= -\frac{5}{2}e^{x+y}
\end{aligned}$$

Example $(Dx^2 - 2DxDy + Dy^2)u = 12xy$

$$\begin{aligned}
(Dx - Dy)^2u &= 12xy \\
u &= \frac{1}{(Dx - Dy)^2} 12xy = \frac{1}{Dx^2} \cdot \frac{1}{\left(1 - \frac{Dy}{Dx}\right)^2} 12xy = \frac{1}{Dx^2} \left(1 - \frac{Dy}{Dx}\right)^{-2} 12xy
\end{aligned}$$

Taylor expansion at 0

$$\begin{aligned}
(1-x)^{-2} &= 1 + 2x + C_3 x^2 + \dots \\
\left(1 - \frac{D_y}{D_x}\right)^{-2} &= 1 + 2 \frac{D_y}{D_x} + C_3 \frac{D_y^2}{D_x^2} + \dots \\
\left(1 - \frac{D_y}{D_x}\right)^{-2} \cdot 12xy &= 12xy + 2 \left(\frac{1}{D_x} D_y\right) \cdot 12xy + C_3 \frac{1}{D_x^2} D_y^2 12xy + \dots \\
2 \frac{1}{D_x} \cdot D_y 12xy &= 2 \frac{1}{D_x} \cdot 12x = 26x^2
\end{aligned}$$

Then

$$u_p = \frac{1}{Dx^2} (12xy + 12x^2) = x^4 + 2x^3y$$

Example $u_{xx} - u_{yy} = xy - x$

$$\begin{aligned}
u &= \frac{1}{Dx^2 - Dy^2} (xy - x) \\
&= \frac{1}{Dx - Dy} \cdot \frac{1}{Dx + Dy} (xy - x) \\
u &= \frac{1}{Dx^2} \cdot \frac{1}{1 - \left(\frac{Dy}{Dx}\right)^2} (xy - x) = \frac{1}{Dx^2} \cdot \left(1 + \frac{Dy^2}{Dx^2} + \left(\frac{Dy^2}{Dx^2}\right)^2 + \dots\right) (xy - x) \\
&= \frac{1}{Dx^2} (xy - x) = \frac{1}{6} x^3 (y - 1)
\end{aligned}$$

Example $u_{xx} + u_{yy} = x^2 + xy$

$$\begin{aligned}
(Dx^2 + Dy^2)u &= x^2 + xy \\
u &= \frac{1}{Dx^2} \frac{1}{1 + \frac{Dy^2}{Dx^2}} (x^2 + xy) \\
&= \frac{1}{Dx^2} (x^2 + xy)
\end{aligned}$$